

Kerem Yildirim

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EDUCATION

- **M.Sc. Informatics**

Technical University of Munich - Munich - Germany
Thesis: Probabilistic Object Detection and Reconstruction from a single RGB-D frame
2019 - 2022

- **B.Sc. Computer Science and Engineering**

Sabanci University - Istanbul, Turkey
Thesis: Plant Disease Classification
2015 - 2019

EXPERIENCE

- **Deep Learning Engineer - Ramblr.ai**

Built auto-annotation pipelines with state-of-the-art VLMs and vision models, leading to **70–95%** reduced annotation effort – best case needing only 4–5 out of 120 videos. Architected the active learning framework from scratch for continuous model improvement. Trained and deployed models across projects spanning Robotics, Automotive, and Agriculture using Ray.
March 2024 - Present, Germany

- **AI Consultant - Netlight Consulting**

Built ML serving pipeline for a leading medical company, packaging 3D scan segmentation and classification into automated reporting – **12× speedup** (120s to 10s per scan). Architecture adopted internally as a reusable project template. Led 2 internal **Triton Inference Server** workshops (~20 engineers).
December 2022 - March 2024, Germany

- **Machine Learning Engineer - Isarsoft**

Trained and deployed real-time person detection and multi-object tracking on NVIDIA Jetson edge devices using **Triton Inference Server** and **TensorRT**, enabling real-time data analytics.
October 2021 - October 2022, Germany

- **Computer Vision Engineer - Research Neutron Source FRM II**

Ported Python neutron image segmentation pipeline to **C++/CUDA**: **100× speedup** (200ms to 2ms per image). Enabled processing of terabyte-scale datasets previously infeasible.
October 2020 - October 2021, Germany

LANGUAGES

- Turkish - Native
- English - C1
- German - B1

SKILLS

- C/C++, Python, CUDA, OpenMP, MATLAB, $\LaTeX$
- Ray, PyTorch, OpenCV, TensorRT, Triton Inference Server, MLflow, Hugging Face Transformers
- Linux, Docker, CMake, Git, ROS, AWS, Azure

PROJECTS

- **Show2Instruct**

Enhancing LLMs with spatial understanding to validate building plans against regulations. Ongoing Ramblr.ai research initiative in collaboration with TU Clausthal, TU Rostock and Neobim.

Python, Blender

- **Monte Carlo Dropout for Object Detection on Point Clouds**

Integrated Monte Carlo dropout into a 3D object detector on ScanNet to estimate predictive uncertainty and improve detection performance.

Python, PyTorch

- **KinectFusion**

Implemented the KinectFusion algorithm (Newcombe et al.) for real-time 3D scene reconstruction from depth images.

CUDA, C++, CMake, Eigen, Ceres

PUBLICATIONS

- BARISTA: A Multi-Task Egocentric Benchmark for Compositional Visual Understanding - In Submission
- One Table to Count Them All: Parallel Frequency Estimation on Single-Board Computers - EuroPar 2019
- Plant Identification with Deep Learning Ensembles - ExpertLifeCLEF 2018

HOBBIES

- Drumming, Specialty Coffee (founder of office coffee club), Diving, Boulderling, Skiing